



CMD + EASE

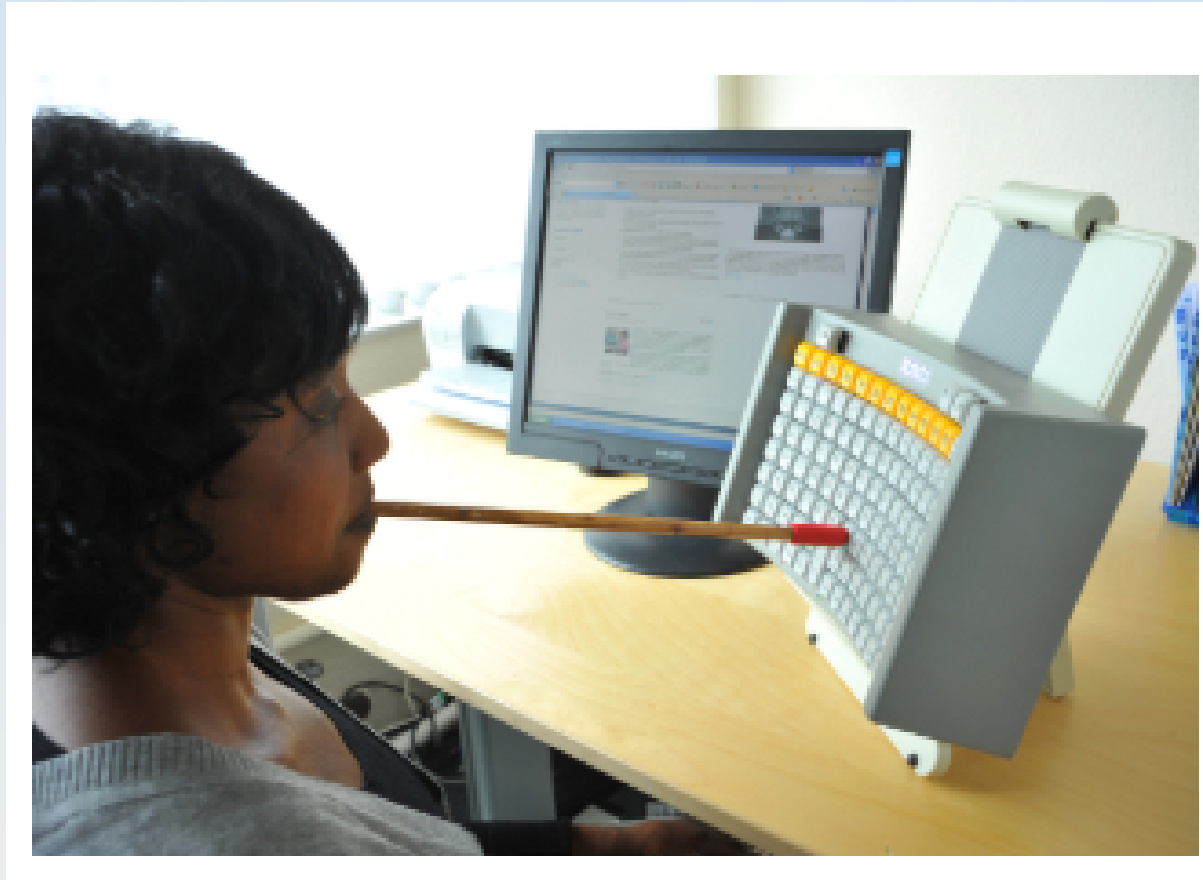
Team Good Ultrons:

Vinayagasubramanian Lakshmisurya

Agrawal Anushka

Louis Au Minove

PROBLEM STATEMENT



Current Market:

- Ergonomic Keyboards
- Specialist Keyboards
- Custom Shortcuts
- Larger Keys
- Sticky Keys
- Continuous pressing of “Tab”
- Separate accessibility mode

Weaknesses:

- Static solutions
- High price point
- Manual Setup



CMD + EASE



1. Real-time detection

- Accessibility mode is either activated when unstable cursor activity is detected or when the user chooses to do so

2. Adaptive system

- On-screen buttons are mapped onto keyboard keys to reduce reliance on the cursor

3. System prompts user before switching to accessibility mode

- Reduced chances of false-positive disruptions

4. Intuitive UI

- No additional features/extensions to be bought or learnt

DEMONSTRATION

INCLUSIVE SOLUTION FOR ALL

- Situational and progressive change
- Different types and level of motor disabilities
- Caters to everyone without heavy cost



TECH STACK

Mock website (Online shopping Platform)

- UI --- HTML, JS, CSS

Accessibility system

- Python
- Web driver --- Selenium
- UI --- HTML

Cursor stability detection (Machine Learning)

- RandomForest model
- Dataset: Age-related changes in motor planning for prior intentions (mouse tracking reach-to-click task)

ROAD AHEAD

STAGE 1

- Prototyping
- Data collections and training Machine Learning model for improved pattern recognition and prompting
- Testing on mockup website
- User control preferences “Ask me first”

STAGE 2

- Implementation as a browser extension
- A/B testing for adaptive UX
- Improved detection of buttons on various types of websites
- Improvement for context based mapping

STAGE 3

- Implementation onto other web architectures
- Standardisation and implementation to other website formats
- Detection based on user’s baseline and not average population data

STAGE 4

- Train Agentic AI to map keys based on severity of tremors
 - Severe tremors will result in mapped keys to be spaced apart (every 3 keys will be used as opposed to adjacent keys)